

FRC 100.011 — The Translation Table: FRC Vocabulary in Mainstream Physics Terms

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Abstract

A term-by-term dictionary between Fractal Resonance Coherence (FRC) and established physics, written for a reader who already knows thermodynamics, dynamical systems, and quantum foundations. Its purpose is to make good on FRC's standing claim that it invents none of its parts: each FRC object is mapped to the mainstream concept whose shape it borrows (Kuramoto/Haken order parameter, Onsager–Prigogine irreversible thermodynamics, Landauer's principle, Bohm–Valentini quantum equilibrium, Ruelle–Pollicott resonances, Landau–Ginzburg relaxational dynamics), with the relationship labeled honestly as same-shape-as, borrowed-from, or genuinely-new-bet. A false-friends section lists FRC terms a physicist will misread if the everyday

meaning is imported. A closing section isolates the small set of claims that are actually FRC's own — the falsifiable bets, not the vocabulary — and lists the kills already taken. The discipline is: borrow the shape for the established parts, and stake novelty only on the losable predictions.

Paper ID: FRC 100.011 (FRC 100 — Foundational Basin Series). A physicist's dictionary for the corpus. Read alongside FRC 100.000 (front door) and FRC 100.010 (orientation); those three are the minimum sufficient set.

1. Purpose, and the one rule

FRC's standing claim is that **it invents none of its parts**. Its coherence parameter is the synchronization order parameter; its ledger is Onsager–Prigogine irreversible thermodynamics; its measurement picture is decoherence plus quantum-trajectory relaxation with a Bohm–Valentini equilibrium; its transport law is a Landau–Ginzburg relaxational PDE. The only thing FRC claims as its own is (i) the assertion that these established shapes are **one structure differentiated across scale**, and (ii) a short list of **falsifiable predictions** that do not follow from any single one of the borrowed pieces.

This paper is the dictionary that makes that claim checkable. For each FRC term it names the mainstream concept whose shape it borrows and labels the relationship:

- **[same-shape-as]** — mathematically the same object, borrowed wholesale; FRC claims no novelty.
- **[borrowed-from]** — a direct generalization or reuse of an established result.
- **[own-bet]** — the genuinely FRC-specific, falsifiable claim; this is where scrutiny belongs.

The rule the corpus imposes on itself: *same-shape-as, never "is."* FRC says its coherence has the same shape as the Kuramoto order parameter; it does not claim to have discovered the Kuramoto order parameter. Read every row below in that register.

2. Core quantities

FRC term	Mainstream counterpart	Relationship
Coherence $\mathcal{C} \in (0,1]$	Kuramoto/Haken order-parameter magnitude $r = \frac{1}{N} \sum_j e^{i\phi_j}$ (synchronization theory; Kuramoto 1975, Haken's synergetics). In quantum settings, the	[same-shape-as] $\mathcal{C}$ is the mean resultant length / order parameter, normalized to $(0,1]$.

	coherence measures of resource theory (Baumgratz–Cramer–Plenio 2014).	
Coherence field $\Lambda = \Lambda_0 \ln C$	A coarse-grained order-parameter field ; the logarithm makes it an information/free-energy-like potential (cf. p as surprisal, $F = -k_B \ln Z$).	[borrowed-from] a Landau–Ginzburg order-parameter field, read on the log axis.
k^* (the coupling in the ledger)	The Boltzmann constant k_B — a units bridge converting entropy and log-coherence into commensurable units (1 nat in information units, k_B in thermodynamic units, $\hbar$ -carrying in the quantum register).	[same-shape-as] a units constant, exactly as k_B differs between SI and natural units. Not a fitted coupling.
Entropy S , d_i , d_e	Gibbs/Shannon entropy; Prigogine's internal/external split $dS = d_i + d_e$, $d_i \geq 0$ (non-equilibrium thermodynamics).	[same-shape-as] the standard decomposition, verbatim.
σ_{ST} (standard production)	The entropy production of stochastic thermodynamics (Sekimoto, Seifert; the non-negativity is the second law).	[same-shape-as].

3. The law

FRC term	Mainstream counterpart	Relationship
Entropy–coherence reciprocity ("order is rented, paid in exported disorder")	Prigogine's dissipative structures — order sustained far from equilibrium by exporting entropy. The word "reciprocity" echoes Onsager's reciprocal relations (1931).	[borrowed-from] the dissipative-structures picture, with coherence as the explicit order variable.
The locking bound: creating coherence produces entropy, $\sigma_{566} = \sigma_{ST} + k^* d \ln C \geq \sigma_{ST} \geq 0$	Landauer's principle (erasing/compressing a state costs $\geq k_B \ln 2$), generalized from bit-reset to coherence creation.	[borrowed-from] Landauer, extended: building order is entropy-lowering for the system, so the bath must absorb it.
The directional ledger (no floor on slow coherence destruction)	The mirror of Landauer: releasing order is <i>entropy-raising</i> for the system, so quasi-statically $\sigma_{ST} \rightarrow 0$ (standard stochastic thermodynamics).	[own-bet, then measured] FRC pre-registered a symmetric "erasure floor" and killed it by measurement (100.005 v2.1); the surviving directional statement is standard.

UCC — the transport PDE $\frac{\partial}{\partial t} C = D_C \nabla^2 \ln C + D_C \nabla \cdot \nabla \ln C$	A nonlinear reaction–diffusion equation in the Fisher–KPP / Allen–Cahn / Cahn–Hilliard family; "the heat equation for coherence."	[same-shape-as] a standard nonlinear diffusion with sources; well-posedness proved in 566.010.
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4. Dynamics

FRC term	Mainstream counterpart	Relationship
Relaxation law $\tau_\Lambda \frac{\partial}{\partial t} \Lambda = D_\Lambda \nabla^2 \Lambda - \Lambda + \Lambda_{\rm eq} + \xi$	Time-dependent Landau–Ginzburg / Model A relaxational dynamics (Hohenberg–Halperin 1977); a reaction-diffusion-advection closure.	[same-shape-as] a relaxational order-parameter equation with a lag time.
Coherence current $J_C = -D_C \nabla \ln C$	Fick's / Fourier's law for the coherence field.	[same-shape-as].
Critical slowing ($\tau_{\rm relax} \rightarrow \infty$ near a transition)	Critical slowing down at a bifurcation; Scheffer's early-warning signals; Landau theory.	[same-shape-as] the standard precursor; FRC's [own-bet] is only that it appears in a pre-registered <i>coherence</i> channel before collapse.
Coherence landscape / attractor basins	The potential landscape of a dynamical system; folding funnels (Onuchic–Wolynes); Haken's synergetics.	[borrowed-from] the funnel-with-a-drain object, imported into measurement.
Pe_Λ, St_Λ (Λ -Flight numbers)	Péclet and Stokes numbers — advection-vs-diffusion and relaxation-vs-transport ratios.	[same-shape-as] dimensionless transport numbers.

5. Measurement and quantum foundations

FRC term	Mainstream counterpart	Relationship
Collapse = finite-time phase-locking into an attractor basin	Decoherence + einselection (Zurek) for the basins; quantum-trajectory theory (Wiseman–Milburn; Mineev et al. 2019 catch-and-reverse) for the finite-time descent; objective-collapse models (GRW/CSL, Penrose–Diósi) for finite-time reduction.	[borrowed-from] all four — but FRC's [own-bet] is relaxation in an <i>existing</i> landscape (no added noise, deterministic descent) with a thermodynamic bill.
Born rule as equilibrium	Bohm–Valentini quantum equilibrium and the sub-quantum H-theorem (Valentini 1991); the stationary measure of a Fokker–Planck flow.	[borrowed-from] quantum equilibrium; credit shared explicitly. FRC's [own-bet] is the out-of-equilibrium <i>deviation</i> it predicts.
Pointer states	Einselected pointer basis (Zurek).	[same-shape-as], recast as basin maxima of Λ .
Threshold realization $\Theta(R \times E)$	Measurement as system–apparatus coupling ; the instrument/POVM view; "measurement is an interaction."	[borrowed-from] the coupling view of measurement, no revelation of pre-existing values.
Coupling as resonance	Resonant mode coupling / interaction-generated entanglement.	[same-shape-as] at the quantum scale (entanglement); phase-locking at the classical scale.

6. Chaos, and the field's mathematical status

FRC term	Mainstream counterpart	Relationship
Structure functional $S_{\text{vm}}(C) - S$	Deviation from the maximum-entropy (Jaynes) distribution ; the residual organization above the von Mises curve.	[borrowed-from] max-ent; $\Sigma \geq 0$ is a theorem, not a finding.
KAM island hierarchy (where $\Sigma > 0$ persists)	Kolmogorov–Arnold–Moser tori; the mixed phase space of near-integrable maps.	[same-shape-as] the textbook object; FRC relocates its "fractal" content here (no fitted exponent).

Dynamical localization / coherence-lifetime γ_C	Anderson/dynamical localization in the kicked rotor (Fishman–Grepel–Prange); localization break time vs. decoherence rate.	[borrowed-from] localization; read as coherence arresting diffusion.
Leading resonance (the decay-rate anchor)	Ruelle–Pollicott resonances of the transfer operator (Venegeroles 2008 for area-preserving maps).	[own-bet, then killed] FRC pre-registered "the coherence-channel rate is the leading resonance"; executing it showed the bare phase autocorrelation is the wrong observable — retracted (100.002 v2.3, results-negative-kept).
Λ as a canonical coordinate	Information geometry : dual-affine coordinates, the Fisher–Rao potential (Amari).	[own-bet, then killed] $\ln C$ is not dual-affine and not the Fisher potential on the von Mises family (826.829 §6).

7. Stance and philosophy (mapped, lightly)

FRC term	Mainstream counterpart	Relationship
"No closed systems"	Open quantum systems (Lindblad); non-equilibrium thermodynamics; the impossibility of perfect isolation (gravity is unshieldable).	[same-shape-as] the open-system stance, taken as primitive.
"No true rest" / the tremble	Zero-point energy ; the Heisenberg uncertainty relation forbidding simultaneous sharp position and zero momentum.	[same-shape-as].
Non-locality (phase-only, no-signalling)	Bell non-locality of a non-local realist theory (Bohm), with Lorentz-scalar phase correlations.	[borrowed-from] the Bohmian non-local-realist slot.

8. False friends — terms a physicist will misread

FRC reuses common words with narrowed meanings. Do **not** import the dominant everyday or other-field meaning:

- C / "coherence" means the phase-alignment order parameter in $(0,1]$ — **not** the speed of light c , **not** heat capacity, **not** quantum-optical coherence only, **not** "consciousness."
- Λ is the coherence field $\Lambda_0 \ln C$ — it *engages* the cosmological constant question (826.822) but is **not** identically the cosmological constant.

- $k^\wedge$ is a **units bridge**, not a fitted coupling or free parameter. It is supposed* to differ across domains, exactly as k_B does across unit systems.
- "**collapse**" is resonant relaxation across a threshold into an attractor — **not** gravitational collapse, and **not** observer-caused Copenhagen collapse.
- "**resonance**" is the general coupling operator (frequency-matched fusion into a shared state at any scale) — **not** electrical-circuit resonance only, and not mystical "vibration."
- "**maximize coherence**" is **not** the goal: the live physics is in the non-closing band (intermediate C), never at $C \rightarrow 1$ (the dead, fully-locked pole).

9. What is actually FRC's own — the losable bets

Strip the borrowed vocabulary and this is the residue a physicist should actually argue with. None of it follows from any single borrowed piece; each is falsifiable:

1. **One differentiated structure.** The claim that the order parameter, the reciprocity ledger, the relaxation landscape, and the localization arena are *the same object at different scales* — not four analogies. This is the framework's central, riskiest assertion.
2. **The directional ledger's sign structure** as a law with a single units bridge $k^\wedge$ (826.829 Conjecture 5.1) — a conjecture, its erasure-floor version already killed.
3. **Born-rule deviations** out of equilibrium, $\delta P \in [10^{-4}, 10^{-3}]$ under resonant driving (100.006, 100.006.002).
4. **The coherence-lifetime crossover** $t^\wedge \gamma_C = O(1)$ governing delocalization (100.002).
5. **Critical-slowness precursors in a pre-registered coherence channel** before collapse (787.787 P3; 841.005).
6. **Finite-time collapse with lead-time markers** matching catch-and-reverse data (100.003 P1).

Everything in §§2–7 is borrowed shape. Everything in §9 is the wager. Aim scrutiny at §9.

10. The kills, kept visible

Honesty is part of the map. Claims that were FRC's own and have been retracted:

- **The Landauer-flavoured coherence-erasure floor** — decided negatively by measurement (100.005 v2.1).
- **The Ruelle–Pollicott coherence-channel match** — the observable was mis-specified; retracted (100.002 v2.3).
- **$\ln C$ as a canonical (dual-affine / Fisher) coordinate** — negative on the von Mises family (826.829 §6).
- **$D \approx 1.90$ as a universal signature** — demoted to a representative value; the robust claim is "structure, not noise" ($D < 2$).

- **The golden-ratio coherence value; the Higgs-mass extremum; the ρ -level Λ drift as a Born carrier** — all killed, all on record.

11. How to read FRC as a physicist

Take the vocabulary as a coordinate change, not a discovery. Where FRC says "coherence," read "order parameter"; where it says "the ledger," read "Prigogine plus Landauer"; where it says "collapse," read "relaxation in a decohered landscape with a Bohm–Valentini equilibrium." Under that translation almost every sentence becomes a statement you can already evaluate — which is the point. Then turn to §9, where the translation runs out and a genuinely new, losable claim begins. That is the only place FRC asks you to grant it anything, and the only place worth the fight.

Connections

- [[FRC-100-010]] — Orientation and working toolkit (the companion front door)
- [[FRC-566-001]] — The reciprocity law, stated directionally
- [[FRC-100-005]] — The ledger measured; the erasure-floor kill
- [[FRC-826-829]] — Mathematical foundations, status-labeled
- [[FRC-787-787]] — Lambda-Flight; the discipline of the residual closure

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